

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1516

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects. (BL4)*

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I Y SRAVAN KUMAR (192472289) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____ SRAVAN _____

Assessment Tasks

Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.

8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.
9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.
10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

1. Unified Big Data Architecture for a Multinational Retail Organization

Problem

A multinational retail organization collects customer information from **online websites, mobile applications, physical stores, payment systems, and loyalty programs**. The main problem is that customer data is stored separately in different systems. This makes it difficult to understand customer behavior and provide personalized services.

Proposed Solution

A unified Big Data architecture can combine data from all channels into a centralized analytics platform.

Architecture

Online Store + Mobile App + Physical Stores + Payment Systems

↓

Kafka / Data Ingestion

↓

Data Lake / HDFS / Cloud Storage

↓

Spark Batch + Spark Streaming

↓

Customer Analytics + Machine Learning

↓

Recommendation Engine

↓

Data Warehouse

↓

Power BI / Tableau

Important Components

1. Data Collection:

Collect customer purchases, website clicks, product searches, mobile interactions, and store transactions.

2. Data Ingestion:

Apache Kafka can collect real-time customer events from different countries and regions.

3. Data Storage:

A Data Lake or HDFS can store large amounts of structured and semi-structured data.

4. Data Processing:

Apache Spark can clean, transform, and combine customer information.

5. Customer Analytics:

The system can identify:

- Frequently purchased products
- Customer preferences
- Purchase frequency
- Customer location
- Browsing behavior

6. Personalized Recommendations:

Machine learning techniques such as **Collaborative Filtering** and **Content-Based Filtering** can recommend products.

7. Real-Time Analytics:

Spark Streaming or Flink can analyze customer activity immediately.

Benefits

- Provides a single view of the customer
- Improves personalized recommendations
- Increases customer engagement
- Supports real-time decision-making
- Reduces data duplication
- Improves sales and customer satisfaction

Conclusion

A unified Big Data architecture allows the organization to combine online and offline customer information and use it for **real-time analytics and personalized recommendations**.

2. Distributed Computing for Large-Scale Scientific Research

Problem

Scientific research produces huge datasets from **satellites, scientific experiments, simulations, laboratories, sensors, and research instruments**. Processing such datasets using a single computer can take a very long time.

Distributed Computing

Distributed computing divides a large task into smaller tasks and processes them across multiple computers.

Centralized vs Distributed Processing

Centralized System:

Large Dataset → One Powerful Computer → Result

Distributed System:

Large Dataset → Split into Parts → Multiple Computers → Parallel Processing → Combined Result

Technologies

- Hadoop
- Apache Spark
- Distributed File Systems
- Cloud Computing

- GPU clusters

How Parallel Processing Helps

Suppose a dataset contains **1 TB of scientific information**.

Instead of one computer processing the entire dataset, it can be divided among several machines.

For example:

1 TB → 10 Nodes → 100 GB per Node

All nodes process their portions at the same time.

Advantages

1. Faster Processing:

Multiple computers work simultaneously, reducing processing time.

2. Scalability:

More machines can be added when the dataset increases.

3. Fault Tolerance:

Distributed systems can replicate data, so failure of one machine does not necessarily cause complete data loss.

4. Large Data Processing:

Datasets larger than the capacity of a single machine can be processed.

5. Resource Utilization:

Computing resources from multiple machines are used together.

Example

Scientific applications such as climate modelling can divide geographical regions among different processing nodes.

Conclusion

Distributed computing is more effective than centralized processing for large-scale scientific research because **parallel processing improves speed, scalability, and reliability**.

3. Fraud Detection Framework for a Digital Payment Platform

Problem

Digital payment platforms process millions of transactions through **UPI, mobile wallets, debit cards, credit cards, and online banking**. Fraudulent transactions need to be detected quickly.

Proposed Framework

Transaction → Kafka → Stream Processing → Feature Extraction → ML Model → Risk Score → Alert/Block

Components

1. Data Collection

The system collects:

- Transaction amount
- Customer ID
- Location
- Device information
- Time
- Merchant
- Previous transaction history

2. Real-Time Data Ingestion

Apache Kafka can continuously receive transaction events.

3. Stream Processing

Apache Flink or Spark Streaming processes transactions in real time.

The system can immediately check whether a transaction is normal or suspicious.

4. Feature Engineering

Important features can include:

- Transaction amount
- Transaction frequency
- Location
- Device
- Time

- Previous spending behavior

5. Machine Learning

Models such as:

- Random Forest
- XGBoost
- Logistic Regression
- Neural Networks

can classify transactions.

6. Risk Score

For example:

Transaction	Risk	Action
Normal	10%	Allow
Suspicious	60%	Verify
Highly suspicious	95%	Block/Alert

Why Streaming Helps

Traditional batch processing may detect fraud after several hours. Streaming processing can detect suspicious activity **within seconds**.

Why Predictive Models Help

Machine learning can learn from historical fraudulent transactions and identify patterns that may not be detected using simple rules.

Benefits

- Faster fraud detection
- Reduced financial losses
- Improved customer security
- Real-time alerts
- Continuous model improvement

Conclusion

Combining **stream processing and predictive machine learning** provides fast and accurate fraud detection for digital payment platforms.

4. Storing and Processing Multimedia Data in Big Data

Problem

Multimedia data includes **images, audio, and video**. These files are usually much larger than normal text or numerical data. They also require high storage capacity and processing power.

Major Challenges

1. Huge Volume:

Videos and high-resolution images consume large amounts of storage.

2. High Processing Requirements:

Processing video and audio requires significant CPU/GPU resources.

3. Different Formats:

Examples include:

- JPG
- PNG
- MP4
- AVI
- MP3
- WAV

4. High Data Transfer:

Large multimedia files require high network bandwidth.

5. Metadata Management:

Information such as location, timestamp, device, and file type must also be managed.

Suitable Technologies

Storage

- Hadoop HDFS
- Cloud Object Storage
- Amazon S3
- Azure Blob Storage
- Google Cloud Storage

Object storage is particularly suitable for large multimedia files.

Processing

- Apache Spark
- Hadoop
- GPU computing
- TensorFlow
- PyTorch

Databases

Metadata can be stored in:

- MySQL
- PostgreSQL
- MongoDB
- Elasticsearch

Efficient Management

Large files can be:

- Compressed
- Partitioned
- Stored in distributed storage
- Processed in parallel
- Replicated for reliability

Example

Video → Cloud Storage → Metadata Database → Spark/GPU Processing → AI Model → Analytics

AI models can be used for:

- Image recognition
- Video analysis
- Speech recognition
- Object detection

Conclusion

BigData technologies such as **HDFS, cloud object storage, Spark, and GPU processing** provide scalable and efficient management of multimedia data.

5. Disaster Recovery and Business Continuity for Cloud Big Data

Problem

A cloud-based Big Data platform may face failures caused by **hardware problems, software errors, cyberattacks, network failures, or natural disasters**. Therefore, a disaster recovery strategy is required.

Proposed Strategy

Primary System → Replication → Backup/Secondary System → Automated Failover

Important Components

1. Data Replication

Data should be replicated across multiple servers or locations.

For example:

Primary Data → Replica 1 → Replica 2

If one copy fails, another copy is available.

2. Backup

Regular backups should be created and stored separately.

Types include:

- Full backup
- Incremental backup
- Snapshot

3. Redundancy

Important services should have multiple instances.

Instead of:

One Server

use:

Server 1 + Server 2 + Server 3

4. Automated Failover

If the primary server fails, the system automatically switches to another available server.

5. Multi-Region Architecture

Critical cloud applications can maintain copies in different geographical regions.

Example:

India Region → Singapore Region → Backup Region

6. Recovery Objectives

RTO – Recovery Time Objective:

Maximum acceptable time to restore the service.

RPO – Recovery Point Objective:

Maximum acceptable amount of data that can be lost.

7. Monitoring

Automated monitoring should detect:

- Server failure
- Storage failure
- Network problems
- Application errors

Benefits

- High availability
- Reduced downtime
- Data protection
- Faster recovery
- Continuous business operations

Conclusion

Redundancy, replication, backups, monitoring, and automated failover make a cloud Big Data platform more resilient and reliable.

6. Data Quality Issues in Large-Scale Analytics

Problem

Large analytics platforms collect data from many sources. This can create problems such as **inconsistency, duplication, missing values, and incorrect information**.

Major Data Quality Problems

1. Inconsistency

The same information may have different formats.

Example:

India
IND
IN

This can create incorrect analysis.

2. Duplication

The same customer may appear multiple times.

Example:

Customer ID 101 → Ravi Kumar
Customer ID 101 → Ravi Kumar

Duplicates can affect calculations.

3. Missing Values

Some records may not contain required information.

Example:

Customer	Age	Location
Ravi	25	India
Arun	—	India

Missing data can reduce analytical accuracy.

4. Incorrect Data

Wrong values can produce incorrect results.

Governance Strategies

1. Data Validation:

Check whether incoming data follows predefined rules.

2. Data Cleaning:

Remove duplicates and correct incorrect values.

3. Standardization:

Use common formats for dates, names, locations, etc.

4. Data Quality Rules:

Define acceptable values and formats.

5. Data Governance:

Define ownership, responsibilities, standards, and access policies.

6. Metadata Management:

Maintain information about data sources, formats, and meaning.

7. Data Monitoring:

Continuously monitor data quality.

Benefits

Good data quality provides:

- More accurate reports
- Better ML predictions
- Reliable decision-making
- Reduced errors
- Better customer analytics

Conclusion

Data governance and quality management are essential because **poor-quality data can directly produce poor analytical results.**

7. Scalable Recommendation System for an Online Streaming Service

Problem

An online streaming service has millions of users watching movies, TV shows, and videos. The system needs to recommend suitable content to each user.

Proposed Architecture

User Activity → Kafka → Data Lake → Spark → ML Recommendation Model → Recommendation API → User

Data Collected

- Videos watched
- Watch duration
- Likes/dislikes
- Search history
- Ratings
- Skipped videos
- Genres
- Viewing time
- Device information

1. Data Ingestion

Apache Kafka collects user activities in real time.

2. Data Storage

A Data Lake stores historical viewing information.

3. Distributed Processing

Apache Spark processes large amounts of user behavior data.

4. User Behavior Analysis

The system identifies:

- Favorite genres
- Frequently watched actors
- Preferred viewing time
- Recently watched content

5. Recommendation Methods

Collaborative Filtering:

Recommends content based on similar users.

Content-Based Filtering:

Recommends content with similar characteristics to previously watched content.

Hybrid Recommendation:

Combines both approaches.

6. Caching

Redis can store frequently requested recommendations for faster response.

7. Scalability

Kafka partitions and Spark distributed processing allow the system to handle millions of users.

Benefits

- Personalized content
- Faster recommendations
- Higher user engagement
- Longer viewing time
- Improved customer satisfaction

Conclusion

Big Data enables streaming services to analyze massive amounts of user behavior and generate **personalized, scalable, and real-time recommendations**.

8. Role of Cloud Computing in Big Data

Introduction

Cloud computing provides computing, storage, databases, networking, and analytics services through the internet. It has become an important part of modern Big Data systems.

Role of Cloud Computing

Cloud platforms provide:

- Large-scale storage
- Computing resources
- Databases
- Data processing
- Machine learning
- Analytics
- Backup and recovery

Examples include:

- AWS
- Microsoft Azure
- Google Cloud

Advantages

1. Scalability

Organizations can increase or decrease computing resources according to workload.

2. Cost Management

Organizations can avoid purchasing large amounts of physical hardware and can use pay-as-you-go services.

3. Flexibility

Cloud services can support different:

- Databases
- Processing frameworks
- ML tools
- Storage systems

4. High Availability

Cloud providers offer distributed infrastructure and backup options.

5. Faster Deployment

New Big Data infrastructure can be created faster compared with building physical data centers.

Limitations

1. Cost Control

Large-scale workloads can become expensive if resources are not monitored.

2. Security

Sensitive data must be protected through proper access control and encryption.

3. Vendor Lock-In

Depending heavily on one cloud provider can make migration difficult.

4. Network Dependency

Cloud applications depend on network connectivity.

5. Data Privacy

Organizations must carefully manage where and how sensitive data is stored.

Conclusion

Cloud computing provides **scalability, flexibility, and easy access to Big Data infrastructure**, but organizations must carefully manage cost, security, privacy, and vendor dependency.

9. Real-Time Analytics for Transportation and Logistics

Problem

Transportation companies manage large amounts of data from **GPS devices, vehicles, traffic systems, weather services, fuel sensors, and delivery systems**. Real-time analytics can improve transportation efficiency.

Proposed Architecture

GPS + Vehicles + Traffic + Weather → Kafka → Spark/Flink → Analytics/ML → Route Optimization → Dashboard

Data Sources

- GPS location
- Vehicle speed
- Traffic information
- Fuel consumption
- Weather
- Delivery information
- Driver behavior

1. Real-Time Data Collection

GPS and vehicle sensors continuously generate data.

Kafka can collect these events.

2. Stream Processing

Spark Streaming or Flink can analyze the data in real time.

3. Route Optimization

The system can compare:

- Traffic conditions
- Distance
- Road conditions
- Weather
- Delivery deadlines

and identify better routes.

4. Fuel Efficiency

Fuel sensor data can be analyzed to identify:

- Excessive fuel consumption
- Poor driving patterns
- Unnecessary idling
- Inefficient routes

5. Predictive Analytics

Machine learning can predict:

- Delivery delays
- Traffic congestion
- Vehicle maintenance requirements
- Estimated arrival time

6. Dashboard

A dashboard can display:

- Vehicle locations
- Delivery status
- Traffic conditions
- Fuel usage
- Estimated arrival time
- Route alerts

Benefits

- Reduced travel time
- Lower fuel consumption
- Better delivery performance
- Improved fleet utilization
- Faster decision-making

Conclusion

Real-time Big Data analytics helps transportation and logistics companies make **faster and more accurate operational decisions**.

10. Ethical and Legal Challenges in Big Data

Introduction

Big Data allows organizations to collect and analyze huge amounts of personal information. However, improper collection or use of data can create ethical and legal problems.

Major Ethical Challenges

1. Privacy

Organizations may collect personal information without users fully understanding how it will be used.

2. Lack of Consent

Users should be informed about data collection and its purpose.

3. Data Security

Poor security can result in data breaches and unauthorized access.

4. Bias

Machine learning models can produce unfair results if training data contains bias.

5. Transparency

Users may not know how automated systems make decisions about them.

6. Data Ownership

Organizations must determine who can control and use collected data.

Legal Requirements

Organizations must follow applicable data protection and privacy laws based on their country and industry, such as **GDPR** and other relevant national regulations.

Recommended Organizational Policies

1. Data Minimization

Collect only the information that is necessary.

2. User Consent

Obtain appropriate consent before collecting or using personal information where required.

3. Encryption

Protect data during storage and transmission.

4. Access Control

Only authorized employees should access sensitive data.

5. Data Anonymization

Remove or protect identifying information when possible.

6. Audit Logs

Record who accessed or changed important information.

7. Data Retention Policy

Do not keep data longer than necessary.

8. Bias Testing

Regularly evaluate ML models for unfair or discriminatory outcomes.

9. Employee Training

Employees should be trained in privacy, security, and responsible data handling.

Benefits

These policies help organizations:

- Protect users
- Reduce privacy risks
- Meet legal requirements
- Improve data security
- Build customer trust

Conclusion

Big Data should be used responsibly. Organizations need strong **privacy, security, governance, transparency, and compliance policies** to maintain user trust while obtaining value from data.